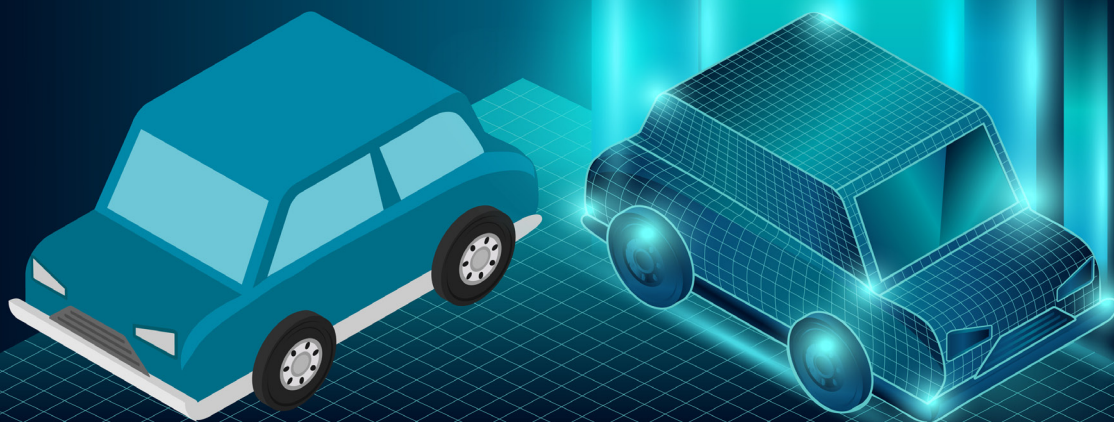


New Smarter And Faster Ways Of CAR MANUFACTURING

Curious about the smarter side of car manufacturing? New tech like virtual tests and digital twins are shaping the way to smarter and faster car production techniques, thus transforming the way cars are built for improved efficiency and speed

digital twin



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The landscape of product development is undergoing a transformative shift, and we are right in the middle of it, especially in the automotive industry. The traditional timelines for creating and launching new vehicles are being dramatically compressed. Companies across the globe are now aiming to bring new products to market in under two years. This acceleration presents formidable challenges, as the complexity of vehicles continues to rise with the integration of cutting-edge technologies and electronic control units (ECUs).

Electric vehicles (EVs) epitomise this shift. Unlike internal combustion engines, which have had decades to evolve, EV technology must be developed and refined

concurrently with product development. This means that every component and system must be rigorously tested and validated before deployment, a process that is both time-consuming and costly.

Moreover, modern vehicles are evolving into multifunctional smart devices, offering a range of services beyond mere transportation. This adds another layer of complexity to their development, as they must now support a variety of software and hardware systems.

To address these challenges, the industry is increasingly turning to virtualisation. By simulating and testing components and systems in a virtual environment, developers can significantly reduce